Multi-Frequency Motion Analysis for Text-to-Motion Retrieval

ABSTRACT

Text-Motion Retrieval (TMR) aims to retrieve 3D motion sequences semantically relevant to text descriptions. However, matching 3D motions with text remains highly challenging, primarily due to the intricate structure of human body and its spatial-temporal dynamics. Existing approaches often overlook these complexities, relying on general encoding methods that fail to distinguish different body parts and their dynamics, limiting precise semantic alignment. To address this, we propose WaMo, a new wavelet-based multi-frequency feature extraction framework. It fully captures **joint**-specific and time-varying motion details across multiple resolutions on body joints, extracting discriminative motion features to achieve fine-grained alignment with texts. WaMo has three key components: (1) Trajectory Wavelet Decomposition decomposes motion signals into frequency components that preserve both local kinematic details and global motion semantics. (2) Trajectory Wavelet Reconstruction uses learnable inverse wavelet transforms to reconstruct original joint trajectories from extracted features, ensuring the preservation of essential spatial-temporal information. (3) Disordered Motion Sequence Prediction reorders shuffled motion sequences to improve learning of inherent temporal coherence, enhancing motion-text alignment. Extensive experiments demonstrate WaMo's superiority, achieving 17.0% and 18.2% improvements in *Rsum* on HumanML3D and KIT-ML datasets, respectively, outperforming existing state-of-the-art (SOTA) methods. Code will be open-sourced upon acceptance.

1 INTRODUCTION

Text-Motion Retrieval (TMR) (Petrovich et al., 2023; Yin et al., 2024; Yu et al., 2024) is an emerging cross-modal retrieval task that has drawn significant attention in recent years. The primary objective is to retrieve semantically relevant 3D motion sequences from a database based on user-provided text queries. However, the inherent complexity of human body structure (e.g., limbs, torso) and their distinct spatial-temporal dynamics makes the robust semantic alignment of 3D motion and text significantly more challenging than conventional cross-modal retrieval tasks (Gorti et al., 2022; Wu et al., 2023; Reddy et al., 2025). Thus, successfully addressing TMR depends on effectively modeling the intricate structure and temporal complexities of the motion modality.

Existing methods often overlook the inherent complexity of 3D motion sequences, relying on conventional coarse-grained methods for representation extraction. Many works (Petrovich et al., 2023; Yang et al., 2024b; Yu et al., 2024; Yin et al., 2024; Li et al., 2025) indiscriminately process the human motion in different parts and moments. They simply regard all joints as a whole and do not consider the variations between moments. **While MoPa (Yu et al., 2024) uses patch-based motion modeling to capture local spatiotemporal structures, it pools joints into different parts to simplify the skeletons, limiting its effectiveness on dense skeletons.** Such methods fail to adequately capture the fine-grained, **joint**-specific dynamics within motions, limiting precise semantic alignment with text descriptions. As illustrated in Figure 1, multi-scale wavelet decomposition of human joint signals displays substantially richer and more discriminative information. It demonstrates the necessity of refined motion signal processing to establish comprehensive correspondence with textual descriptions.
`![A 3D rendered human figure in a blue suit, standing on a grey platform, performing a motion sequence.](61972112770d9c8fb00883057954f885_img.jpg)`

A 3D rendered human figure in a blue suit, standing on a grey platform, performing a motion sequence.

Text Description:
"The person *pulls* something towards him with his right hand, *picks* something up with his left hand, *wipes* it off using *right hand* and *puts it down in his left hand*"

`![A line graph showing the original motion sequence for the left arm (red line) and right arm (blue line) over time. The x-axis represents time from 0 to 150, and the y-axis represents amplitude from 0.2 to 1.0. The graph shows two distinct peaks, one for each arm, corresponding to the actions described in the text.](8d2f4e036a5e0205a1ea3f0b3378f584_img.jpg)

A line graph showing the original motion sequence for the left arm (red line) and right arm (blue line) over time. The x-axis represents time from 0 to 150, and the y-axis represents amplitude from 0.2 to 1.0. The graph shows two distinct peaks, one for each arm, corresponding to the actions described in the text.

(a) Original Human Motion

![A line graph showing the approximation of the original motion sequence at scale 0. The x-axis represents time from 0 to 150, and the y-axis represents amplitude from 0.5 to 2.0. The graph shows a smooth, low-frequency version of the original motion.](131b27192f21ffa3869d8686b7a67448_img.jpg)

A line graph showing the approximation of the original motion sequence at scale 0. The x-axis represents time from 0 to 150, and the y-axis represents amplitude from 0.5 to 2.0. The graph shows a smooth, low-frequency version of the original motion.

![A line graph showing the detail of the motion sequence at scale 1. The x-axis represents time from 0 to 150, and the y-axis represents amplitude from -0.06 to 0.08. A red box highlights a specific region of the graph, corresponding to the 'wipe it off using right hand' action.](6d06e66b7a838f8c809884aeee597a5e_img.jpg)

A line graph showing the detail of the motion sequence at scale 1. The x-axis represents time from 0 to 150, and the y-axis represents amplitude from -0.06 to 0.08. A red box highlights a specific region of the graph, corresponding to the 'wipe it off using right hand' action.

![A line graph showing the detail of the motion sequence at scale 2. The x-axis represents time from 0 to 150, and the y-axis represents amplitude from -0.04 to 0.06. A blue box highlights a specific region of the graph, corresponding to the 'put it down in his left hand' action.](1ac0e11d90bece49015adc89be472a39_img.jpg)

A line graph showing the detail of the motion sequence at scale 2. The x-axis represents time from 0 to 150, and the y-axis represents amplitude from -0.04 to 0.06. A blue box highlights a specific region of the graph, corresponding to the 'put it down in his left hand' action.

(b) Decomposed Motion

Figure 1: **Wavelet decomposition of human motion signals.** For the original human motion trajectory of left and right arms (shown in the top-right subfigure, where the x-axis denotes time and y-axis denotes the amplitude of movements), we perform wavelet decomposition across three distinct scales. The scale-0 waveform preserves the overall structure of the original motion sequence. Compared to the scale-0 waveform, scale 1 highlights the rapid “wipe it off using right hand” movement (as in **red boxes**), while scale 2 captures higher-frequency details of two-handed interactions related to “put it down in his left hand” (as in **blue boxes**). It demonstrates how frequency-specific decomposition reveals the motion-text correspondence.

To address these limitations, we propose a comprehensive approach for processing kinematic information across multiple human joints through multi-frequency feature extraction. In particular, we adopt wavelet decomposition. Alternative methods such as convolutions primarily emphasize high frequencies due to their constrained receptive field (Wang et al., 2020). Prior works on single-modal motion prediction and generation have used Fourier Transform (FT) (Starke et al., 2020; 2022; Wan et al., 2023) or the Discrete Cosine Transform (DCT) (Mao et al., 2019; Chen et al., 2023) for human motion modeling. However, Fourier Transform utilizes a global frequency representation, thereby struggling to capture essential local temporal dynamics. DCT discards high-frequency components, which represent short and abrupt motions that are important in TMR. In contrast, wavelet decomposition effectively captures low-frequency components (Fujieda et al., 2018) while simultaneously preserving localized information (Finder et al., 2024). Technically, we extract low-frequency components to capture long-term movement trends, while simultaneously obtaining high-frequency features to represent short and abrupt motions. This multi-frequency analysis results in discriminative motion features that facilitate fine-grained alignment with textual semantics. Unlike prior methods (shengchuan gao et al., 2025; Feng et al., 2024) that do not distinguish the fine-grained semantics of low- and high-frequency components, we aim to capture both frequency-specific characteristics and their inter-dependencies, facilitating fine-grained alignment with textual semantics. To further improve the feature extraction process, we propose a learnable inverse wavelet transform module, which reconstructs the original joint trajectories from motion features. It ensures that the extractor has effectively captured motion characteristics inherent in the original joint movements.

Moreover, since text-described actions naturally follow a temporal order aligned with motion sequences, the model needs to capture the temporal dynamics of 3D motions. To enhance the understanding of temporal dynamics, we introduce a sequence reordering task. It involves reconstructing the correct order from randomly shuffled motion sequences, which encourages the model to learn

and maintain the temporal dependencies in the motions. As a result, it improves the model's ability to align motion with textual descriptions.

Concretely, we introduce WaMo, a novel wavelet-based multi-frequency feature extraction framework for TMR. Our framework consists of three key components: (1) Trajectory Wavelet Decomposition (TWD) decomposes motion signals into frequency components that preserve both local kinematic details and global motion semantics. By designing a special Intra- and Inter-Frequency Attention module, it can better capture both frequency-specific characteristics and their inter-dependencies. (2) Trajectory Wavelet Reconstruction (TWR) introduces a constraint using learnable inverse wavelet transforms to reconstruct the original joint trajectory from the extracted features. It acts as a regularization term to make sure the motion encoder extracts more pertinent kinematical information. (3) Disordered Motion Sequence Prediction (DMSP) randomly shuffles motion sequences and trains the model to recover their original temporal orders. This strategy explicitly enforces the learning of inherent motion dynamics, improving the alignment between sequential motion dynamics and textual descriptions.

In summary, our key contributions are as follows:

- We introduce WaMo, a novel wavelet-based multi-frequency feature extraction framework. It captures both intra-frequency characteristics and inter-frequency dependencies across multiple scales via wavelet transforms, leading to refined 3D motion representations.
- We propose the trajectory wavelet reconstruction module, using learnable inverse wavelet transforms to reconstruct original human motion trajectories from extracted features. It ensures the preservation of spatial-temporal information within features and enhances the feature robustness and discrimination.
- Extensive experiments on two datasets, HumanML3D (Guo et al., 2022) and KIT-ML (Plappert et al., 2016), validate the superiority of our method. WaMo outperforms existing state-of-the-art methods by substantial margins, achieving 17.0% and 18.2% improvements in *Rsum* on these datasets, respectively.

2 RELATED WORKS

2.1 TEXT-MOTION RETRIEVAL

In recent years, text-motion retrieval has received much attention, which aims to achieve mutual matching between text descriptions and 3D human motions. In particular, TMR (Petrovich et al., 2023) extends the text-to-motion generation model TEMOS (Petrovich et al., 2022) with a contrastive loss. MoPa (Yu et al., 2024) regards XYZ coordinates as RGB pixels to encode 3D motions as 2D images. LAVIMO (Yin et al., 2024) introduces videos as an additional modality to bridge the gap between texts and motions. MGSI (Yang et al., 2024b) formulates text-motion retrieval as a multi-instance multi-label learning problem. MESM (Shi & Zhang, 2024) adopts a large language model to expand coarse text descriptions into fine-grained ones. LaMP (Li et al., 2025) conducts joint training for text-motion retrieval, text-to-motion generation, and motion-to-text captioning. However, they adopt coarse-grained methods to extract motion representations, overlooking the intricate nature of human joints. In contrast, we adopt multi-frequency wavelet transforms to decompose human motions, leading to more detailed information.

2.2 CROSS-MODAL SEMANTIC ALIGNMENT

Cross-modal semantic alignment is a critical yet challenging task that aims to align different modalities in the latent semantic space (Zhou et al., 2021; Wu et al., 2022; Jin et al., 2023; Fu et al., 2024; Feng et al., 2025). DE++ (Dong et al., 2021) learns global and local patterns in latent space and concept space. VT-TWINS (Ko et al., 2022) aligns noisy and weakly correlated multi-modal time-series data using differentiable Dynamic Time Warping. TRM (Zheng et al., 2023) mines phrase-level temporal relationships between videos and sentences. SSN (Yang et al., 2024a) decomposes semantic shifts within modalities to guide cross-modal alignment. CMA (Kim & Hwang, 2025) regularizes the distances between image and text embeddings in the hyper-spherical representation space. Generally, existing methods mainly focus on 2D images or videos. Different from them, we perform cross-modal alignment on complex 3D motions and texts.

![Figure 2: The overview of WaMo. The diagram illustrates the architecture of the model, which consists of three main components: Disordered Motion Sequence Prediction (DMSP), Trajectory Wavelet Decomposition (TWD), and Trajectory Wavelet Reconstruction (TWR). DMSP takes a disordered motion sequence and uses temporal structure learning and a classification layer to predict the correct temporal order. TWD uses a stationary wavelet transform to decompose human joint trajectories into frequency components (low and high). TWR uses a motion decoder and learnable synthesis filters to reconstruct the original motion trajectories from the decomposed components. The model also incorporates a text description (e.g., 'A person walking and helping maintain their balance and support, from holding onto a side rail or wall.') which is processed through a text encoder and similarity learning to provide contextual information for the motion prediction.](2fa4a1bf91d0f34e87c689fbc1211fe3_img.jpg)

The diagram shows the WaMo architecture. At the top left, 'Disordered Motion Sequence Prediction' shows a shuffled sequence of human figures being processed through 'Temporal Structure Learning' (represented by a matrix) and a 'Classification Layer' to restore the temporal order. Below this, 'Disorder' is shown as a shuffled sequence. In the center, 'Trajectory Wavelet Decomposition' shows a 'Human Motion Sequence' being converted to a 'Human Joint Trajectory' (a 3D plot), which is then processed by a 'Stationary Wavelet Transform' using a 'Scaling Function $\phi(t)$ ' and 'Wavelet Function $\psi(t)$ '. This results in 'Wavelet Vectors' for 'Higher Freq.' and 'Lower Freq.' components. These are further processed by an 'Intra-Bin-Frequency Attention Module' to produce 'Intra-Frequency Feature' (Higher Freq.) and 'Inter-Frequency Feature' (Lower Freq.). At the top right, 'Trajectory Wavelet Reconstruction' shows the 'Reconstructed Human Joint Trajectory' being processed by an 'Inverse Stationary Wavelet Transform' using 'Learnable Synthesis Filters' and a 'Motion Decoder' to produce the final output. At the bottom, a 'Text Description' is processed through a 'Text Encoder' to create a 'Text Embedding', which is then used for 'Similarity Learning' and 'Attention Pooling' to enhance the motion prediction.

Figure 2: The overview of WaMo. The diagram illustrates the architecture of the model, which consists of three main components: Disordered Motion Sequence Prediction (DMSP), Trajectory Wavelet Decomposition (TWD), and Trajectory Wavelet Reconstruction (TWR). DMSP takes a disordered motion sequence and uses temporal structure learning and a classification layer to predict the correct temporal order. TWD uses a stationary wavelet transform to decompose human joint trajectories into frequency components (low and high). TWR uses a motion decoder and learnable synthesis filters to reconstruct the original motion trajectories from the decomposed components. The model also incorporates a text description (e.g., 'A person walking and helping maintain their balance and support, from holding onto a side rail or wall.') which is processed through a text encoder and similarity learning to provide contextual information for the motion prediction.

Figure 2: **The overview of WaMo.** We comprehensively model the intricate spatial-temporal dynamics of 3D human motion and establish precise alignment with textual semantics. WaMo consists of three key components: (1) **Trajectory Wavelet Decomposition** (TWD) decomposes motion signals into frequency components that preserve both local kinematic details and global motion semantics. It enables comprehensive multi-scale frequency analysis. (2) **Trajectory Wavelet Reconstruction** (TWR) facilitates accurate reconstruction of original motion trajectories, thereby ensuring robust motion representation learning. (3) **Disordered Motion Sequence Prediction** (DMSP) enhances temporal understanding by training the model to recover correct temporal ordering from shuffled motion sequences, improving the modeling of motion dynamics.

3 PRELIMINARIES

Wavelet decomposition is widely adopted to analyze signals at multiple levels of detail (Mallat, 2002; Guo et al., 2017; Yao et al., 2022; Oka et al., 2025). Unlike Fast Fourier Transform (FFT), which primarily focuses on global frequency representation, wavelet decomposition captures both global and local temporal dynamics. In particular, Discrete Wavelet Transform (DWT) conducts shift-variant decomposition, disrupting the original temporal structure. Therefore, we employ Stationary Wavelet Transform (SWT), which captures multi-scale temporal dynamics while preserving the original temporal structure of signals.

3.1 STATIONARY WAVELET TRANSFORM

Let $x[n]$ be a discrete time signal and $h[k]$, $g[k]$ the low-pass and high-pass filters obtained by scaling function $\phi(t)$ and wavelet function $\psi(t)$, where k specifies the translation. At level s, approximation coefficients a_s and detail coefficients d_s are computed as:

$$a_s[n] = \sum_k h[k] a_{s-1}[n + 2^{s-1}k], \quad d_s[n] = \sum_k g[k] a_{s-1}[n + 2^{s-1}k], \quad (1)$$

with the initialization $a_0[n] = x[n]$. The complete SWT decomposition of level S recursively produces the coefficient set $\{d_1, d_2, \dots, d_S, a_S\}$.

3.2 INVERSE STATIONARY WAVELET TRANSFORM

Let $\tilde{h}[k]$, $\tilde{g}[k]$ be the synthesis filters forming a precise-reconstruction pair with h, g. Starting from a_S and the detail sequences d_s, the reconstructed low-pass signal at level $s - 1$ is:

$$a_{s-1}[n] = \sum_k \tilde{h}[k] a_s[n - 2^{s-1}k] + \sum_k \tilde{g}[k] d_s[n - 2^{s-1}k], \quad (2)$$

After S levels, the original signal is recovered: $\hat{x}[n] = a_0[n]$.

4 METHOD

4.1 OVERVIEW

The overall framework of WaMo is illustrated in Figure 2. Our method employs wavelet transforms to comprehensively analyze the spatial-temporal characteristics of human joint movements and establish robust correspondence with textual semantics. In Section 4.2, we first detail our text encoder. We then present the Trajectory Wavelet Decomposition (TWD) to encode motion sequence signals in Section 4.3, which decomposes motion signals into multi-scale frequency components to fully capture motion dynamics. The Trajectory Wavelet Reconstruction (TWR) is subsequently introduced in Section 4.4, ensuring the preservation of spatial-temporal information. Finally, we present Disordered Motion Sequence Prediction (DMSP) in Section 4.5, where the model learns temporal structures by recovering the correct ordering of shuffled motion sequences.

4.2 TEXT ENCODER

For the textual representation, we adopt the pre-trained DistilBERT (Sanh et al., 2019) to encode text embeddings, following prior works (Petrovich et al., 2023; Yu et al., 2024; Shi & Zhang, 2024). The output of the [CLS] token is projected into the latent semantic space of motion-language modalities. The encoded text embedding is denoted as $t \in \mathbb{R}^D$, where D is the latent dimension.

4.3 TRAJECTORY WAVELET DECOMPOSITION

Given a motion sequence with T frames and J joints in xyz coordinates, we denote it as $M \in \mathbb{R}^{T \times J \times 3}$. To decompose it into frequency components that preserve both local kinematic details and global motion semantics, we adopt learnable SWT (Nason & Silverman, 1995). It provides a time-invariant decomposition and preserves the original temporal structure. SWT is performed along time per joint coordinate, leading to low-frequency and high-frequency vectors $M_{low} \in \mathbb{R}^{T \times J \times 3}$ and $M_{high} \in \mathbb{R}^{S \times T \times J \times 3}$.

$$247 \quad M \xrightarrow{SWT(h_0, g_0)} \{M_{low}, M_{high}\}, \quad (3)$$

where h_0 and g_0 are learnable low-pass and high-pass filters, respectively, and S is the decomposition level. M_{low} and M_{high} are extracted with temporal patterns at distinct scales, capturing both local/global patterns across multiple levels.

4.3.1 INTRA- AND INTER-FREQUENCY ATTENTION

To fully capture comprehensive motion characteristics across diverse temporal scales, we propose the Intra- and Inter-Frequency Attention module. The pipeline is shown in Figure 3. For each frequency component, we first employ specialized 1D convolutions: long-term temporal patterns in M_{low} are captured using large kernels, while short-term variations in M_{high} are extracted via small kernels. The processed features are then flattened and mapped by a multi-layered perceptron (MLP), followed by a Transformer encoder (Vaswani et al., 2017). This yields the intra-frequency representations $\tilde{M}_{low} \in \mathbb{R}^{T \times D}$ for global motion trends and $\tilde{M}_{high} \in \mathbb{R}^{S \times T \times D}$ for local kinematic details.

![Figure 3: Pipeline of the Intra- and Inter-Frequency Attention module. The diagram shows two parallel processing paths for Low Freq and High Freq wavelet components. The Low Freq path uses a 'Conv. with Large Kernel' followed by an 'Intra-Attention Layer' to produce 'Low Freq Feats'. The High Freq path uses a 'Conv. with Small Kernel' followed by an 'Intra-Attention Layer' to produce 'High Freq Feats'. These two are then combined in an 'Inter-Frequency Attention' block to produce the final 'Multi-Freq Feats'.](61374b1b60dadda31f7e83db5cbdde10_img.jpg)

Figure 3: Pipeline of the Intra- and Inter-Frequency Attention module. The diagram shows two parallel processing paths for Low Freq and High Freq wavelet components. The Low Freq path uses a 'Conv. with Large Kernel' followed by an 'Intra-Attention Layer' to produce 'Low Freq Feats'. The High Freq path uses a 'Conv. with Small Kernel' followed by an 'Intra-Attention Layer' to produce 'High Freq Feats'. These two are then combined in an 'Inter-Frequency Attention' block to produce the final 'Multi-Freq Feats'.

Figure 3: The pipeline of the Intra- and Inter-Frequency Attention module. It preserves frequency-specific characteristics while capturing their inter-dependencies.

To integrate multi-scale motion characteristics, we concatenate the intra-frequency features along the feature dimension, forming the combined representation $\tilde{M}_{multi} \in \mathbb{R}^{T \times (S+1)D}$. This fused

feature captures both global motion trends ($\tilde{M}_{low}$) and local kinematic details ($\tilde{M}_{high}$). It is then processed through an MLP and Transformer layer to compute the final inter-frequency feature $\hat{M} \in \mathbb{R}^{T \times D}$, which comprehensively encodes the multi-resolution spatial-temporal dynamics of human motion. Finally, we employ the additive attention pooling (Bahdanau et al., 2015) on $\hat{M}$ to obtain the aggregated motion embedding $m \in \mathbb{R}^D$. To align the text and motion embeddings, following prior works (Petrovich et al., 2023; Yu et al., 2024), we adopt the InfoNCE loss (Oord et al., 2018):

$$\mathcal{L}_{nce} = -\frac{1}{B} \sum_i \left(\log \frac{\exp(S_{ii}/\tau)}{\sum_j \exp(S_{ij}/\tau)} + \log \frac{\exp(S_{ii}/\tau)}{\sum_j \exp(S_{ji}/\tau)} \right), \quad (4)$$

where B is the size of the mini-batch, $S_{ij} = \cos(t_i, m_j)$ is the cosine similarity of the i -th text and j -th motion embeddings within the mini-batch, and τ is the temperature hyperparameter.

4.4 TRAJECTORY WAVELET RECONSTRUCTION

To ensure the encoded intra- and inter-frequency features preserve the original spatial-temporal structures of motions, we further introduce the Trajectory Wavelet Reconstruction (TWR) module. The intra-frequency features $\tilde{M}_{low}$ and $\tilde{M}_{high}$ are transformed through separate MLPs to recover low- and high-frequency vectors $\bar{M}_{low} \in \mathbb{R}^{T \times J \times 3}$ and $\bar{M}_{high} \in \mathbb{R}^{S \times T \times J \times 3}$. Then, we employ inverse SWT (ISWT) to reconstruct the motion sequence, which is denoted as $\bar{M}_{intra} \in \mathbb{R}^{T \times J \times 3}$:

$$\{\bar{M}_{low}, \bar{M}_{high}\} \xrightarrow{ISWT(h_1, g_1)} \bar{M}_{intra}, \quad (5)$$

where h_1 and g_1 are learnable low-pass and high-pass synthesis filters. For the inter-frequency feature $\hat{M}$, we directly input it into an MLP motion decoder to obtain the holistic reconstructed motion $\bar{M}_{inter} \in \mathbb{R}^{T \times J \times 3}$, as it already incorporates cross-frequency correlations. We compare these reconstructed motions to the ground-truth motion M via:

$$\mathcal{L}_{rec} = \mathcal{L}_1(\bar{M}_{intra}, M) + \mathcal{L}_1(\bar{M}_{inter}, M), \quad (6)$$

where $\mathcal{L}_1$ is the smooth L1 loss (Girshick, 2015).

4.5 DISORDERED MOTION SEQUENCE PREDICTION

Inspired by self-supervised Jigsaw Puzzle Solving in image representation learning (Wei et al., 2019; Jing & Tian, 2020; Zhuge et al., 2021), we propose the Disordered Motion Sequence Prediction (DMSP) module to learn temporal structures inherent in motion sequences. However, given the fundamental differences between 2D image/video and 3D human motion, directly applying 2D sequence shuffling techniques (Lee et al., 2017) could indiscriminately rearrange frames across the entire sequence. Such methods could disrupt local kinematic dependencies, resulting in discontinuous and physically impossible actions that confuse the encoder. In contrast, DMSP is specifically designed for the unique dynamics of motion. By dividing sequences into temporally coherent groups and only shuffling a small group of frames, DMSP preserves the local kinematic continuity while forcing the model to reconstruct the global semantic causal order. We provide experiments comparing 2D sequence shuffling techniques (Lee et al., 2017) and our DMSP in Section 5.4.

Technically, we first divide the motion sequence into λ_g temporally coherent groups, where each group contains T/λ_g consecutive frames sharing the same temporal label. Then, we randomly select and shuffle a fraction λ_s of frames to create the shuffled sequence, followed by the TWD module to obtain the disrupted motion feature $\tilde{M} \in \mathbb{R}^{T \times D}$. The model then learns temporal structures by predicting the original and shuffled group labels $g_o, g_s \in \mathbb{R}^T$:

$$p_o = \text{Softmax}(\text{CLS}(\hat{M})), \quad p_s = \text{Softmax}(\text{CLS}(\tilde{M})), \quad (7)$$

where CLS is the classification layer, $p_o \in \mathbb{R}^{T \times \lambda_g}$ and $p_s \in \mathbb{R}^{T \times \lambda_g}$ are the predicted probability distributions of group labels. The training objective is defined as:

$$\mathcal{L}_{dmsp} = \mathcal{L}_{CE}(p_o, g_o) + \mathcal{L}_{CE}(p_s, g_s), \quad (8)$$

where $\mathcal{L}_{CE}$ represents the cross-entropy loss function.

Method	Venue	Text-to-Motion						Motion-to-Text						Rsum [†]
		R@1 [†]	R@2 [†]	R@3 [†]	R@5 [†]	R@10 [†]	MedR _↓	R@1 [†]	R@2 [†]	R@3 [†]	R@5 [†]	R@10 [†]	MedR _↓	
<i>Text-Motion Generation Models:</i>														
T2M	CVPR'22	1.80	3.42	4.79	7.12	12.47	81.00	2.92	3.74	6.00	8.36	12.95	81.50	63.57
TEMOS	ECCV'22	2.12	4.09	5.87	8.26	13.52	173.00	3.86	4.54	6.94	9.38	14.00	183.25	72.58
MotionCLIP	ECCV'22	2.33	5.85	8.93	12.77	18.14	103.00	5.12	6.97	8.35	12.46	19.02	91.42	99.94
MoMask [†]	CVPR'24	2.00	3.58	5.03	7.42	13.17	77.00	1.81	3.58	5.22	7.87	13.31	77.00	62.99
<i>Text-Motion Retrieval Models:</i>														
MoT	SIGIR'23	2.61	4.72	6.90	10.66	17.79	60.00	4.03	5.07	7.43	11.23	17.68	64.25	88.12
TMR	ICCV'23	5.68	10.59	14.04	20.34	30.94	28.00	9.95	12.44	17.95	23.56	32.69	28.50	178.18
MGSI	MM'24	6.61	12.73	17.11	23.91	34.74	24.00	10.61	13.18	19.75	26.00	36.63	22.50	201.27
MESM	MM'24	7.16	12.52	16.70	24.22	35.38	23.00	11.19	13.81	19.59	25.96	35.93	23.25	202.46
LAVIMO	CVPR'24	6.37	11.84	15.60	21.95	33.67	24.00	9.72	13.33	18.73	25.00	36.55	22.50	192.76
MoPa	CVPR'24	<u>10.80</u>	<u>14.98</u>	<u>20.00</u>	<u>26.72</u>	<u>38.02</u>	<u>19.00</u>	<u>11.25</u>	<u>13.86</u>	<u>19.98</u>	<u>26.86</u>	<u>37.40</u>	<u>20.50</u>	<u>219.87</u>
LaMP [†]	ICLR'25	3.65	7.56	10.90	16.56	26.26	30.00	4.51	8.71	11.76	17.34	27.84	30.00	135.09
WaMo (Ours)	-	14.02	17.58	25.51	32.06	42.10	16.00	15.51	16.57	22.74	29.40	41.73	17.00	257.22

Table 1: **Comparison results on HumanML3D (Guo et al., 2022).** The best results are shown in **bold** and the second-best outcomes are underlined. We achieve state-of-the-art results across all metrics. [†] denotes reproduced using official checkpoints.

Method	Venue	Text-to-Motion						Motion-to-Text						Rsum [†]
		R@1 [†]	R@2 [†]	R@3 [†]	R@5 [†]	R@10 [†]	MedR _↓	R@1 [†]	R@2 [†]	R@3 [†]	R@5 [†]	R@10 [†]	MedR _↓	
<i>Text-Motion Generation Models:</i>														
T2M	CVPR'22	3.37	6.99	10.84	16.87	27.71	28.00	4.94	6.51	10.72	16.14	25.30	28.50	129.39
TEMOS	ECCV'22	7.11	13.25	17.59	24.10	35.66	24.00	11.69	15.30	20.12	26.63	36.39	26.50	207.84
MotionCLIP	ECCV'22	4.87	9.31	14.36	20.09	31.57	26.00	6.55	11.28	17.12	25.48	34.97	23.00	175.60
MoMask [†]	CVPR'24	3.69	7.81	12.22	17.47	29.83	22.00	4.55	8.10	11.22	17.19	28.69	22.00	140.77
<i>Text-Motion Retrieval Models:</i>														
MoT	SIGIR'23	6.23	11.07	16.54	23.92	37.15	20.00	10.56	13.49	20.61	27.61	38.04	19.50	205.22
TMR	ICCV'23	7.23	13.98	20.36	28.31	40.12	17.00	11.20	13.86	20.12	28.07	38.55	18.00	221.80
MGSI	MM'24	8.91	16.28	20.87	29.64	40.84	16.00	13.49	16.41	23.54	30.66	43.00	15.50	243.64
MESM	MM'24	9.29	17.05	22.31	29.13	41.02	16.00	12.75	16.41	24.17	32.59	42.88	15.50	247.60
LAVIMO	CVPR'24	10.16	19.92	24.61	34.57	49.80	11.00	15.43	20.12	26.95	34.57	53.32	10.00	289.45
MoPa	CVPR'24	<u>14.02</u>	<u>21.08</u>	<u>28.91</u>	<u>34.10</u>	<u>50.00</u>	<u>10.50</u>	<u>13.61</u>	<u>17.26</u>	<u>27.54</u>	<u>33.33</u>	<u>44.77</u>	<u>13.00</u>	<u>284.62</u>
LaMP [†]	ICLR'25	6.25	12.50	17.76	24.86	42.47	13.00	6.39	14.49	18.89	27.56	44.03	13.00	215.20
WaMo (Ours)	-	18.31	24.82	34.46	43.49	56.75	8.00	19.04	22.41	31.11	37.83	54.04	9.00

342.26 |

Table 2: **Comparison results on KIT-ML (Plappert et al., 2016).** The best results are shown in **bold** and the second-best outcomes are underlined. We achieve state-of-the-art results across all metrics. [†] denotes reproduced using official checkpoints.

5 EXPERIMENTS

5.1 EXPERIMENT SETUP

5.1.1 DATASETS

We employ two widely-adopted datasets to validate the proposed method: HumanML3D (Guo et al., 2022) and KIT-ML (Plappert et al., 2016). **HumanML3D** contains human motions from AMASS (Mahmood et al., 2019) and HumanAct12 (Guo et al., 2020) with additional text descriptions. Following the official split, the training, validation, and test sets have 23384, 1460, and 4380 motions, respectively. Each motion is annotated with 3 text captions on average. **KIT-ML** primarily focuses on locomotion. It is split into 4888, 300, and 830 motions for training, validation, and test sets, respectively. Each motion is associated with an average of 2.1 text captions.

5.1.2 METRICS

We adopt the commonly used Recall at k (R@K), Median Rank (MedR), and Rsum as metrics. R@K is the ratio of queries that retrieve target items in the top-K results. K is set to 1,2,3,5,10. MedR is the median rank of target items. Rsum is the sum of R@K values. Higher R@K, Rsum, and lower MedR indicate better retrieval accuracy.

5.1.3 IMPLEMENTATION DETAILS

In the experiments, we adopt the Adam optimizer (Kingma & Ba, 2015) with a learning rate of 1e-4 and a cosine annealing schedule to train the model. The latent dimension D , decomposition level S , and temperature τ are set to 256, 3, and 0.07, respectively. The group number λ_g and shuffle ratio λ_s are set to 16 and 0.25. For fair comparison with prior works, we follow the same motion pre-processing method proposed in T2M (Guo et al., 2022). Unless otherwise specified, we mainly report text-to-motion retrieval results.

5.2 COMPARISON WITH STATE-OF-THE-ARTS

We compare our method with the following state-of-the-arts (SOTA): **text-motion generation methods**, *i.e.*, T2M (Guo et al., 2022), TEMOS (Petrovich et al., 2022), MotionCLIP (Tevet et al., 2022), MoMask (Guo et al., 2024), and **text-motion retrieval methods**, *i.e.*, MoT (Messina et al., 2023), TMR (Petrovich et al., 2023), MGSI (Yang et al., 2024b), MESM (Shi & Zhang, 2024), LAVIMO (Yin et al., 2024), MoPa (Yu et al., 2024), and LaMP (Li et al., 2025). As shown in Table 1 and 2, our method outperforms prior works by a substantial margin across all metrics on both datasets. Notably, compared to prior SOTA methods, MoPa and LAVIMO, we achieve 17.0% and 18.2% relative improvements on the Rsum metric of HumanML3D and KIT datasets, respectively. These results demonstrate the superior effectiveness of our approach.

5.3 ABLATION STUDIES

5.3.1 MAIN ABLATION STUDIES

In Table 3, we comprehensively assess the effectiveness of the proposed modules. Concretely, when TWD is removed, we directly flatten and map all human joint coordinates in one frame into a token, followed by a Transformer encoder to encode all frames (tokens). To ablate TWR or DMSP, we just remove $\mathcal{L}_{rec}$ or $\mathcal{L}_{dmsp}$, respectively. From the table, it is found that the introduction of each component can bring performance improvements, and the best performance is achieved when all modules are introduced (Row 6).

Row	Setting			HumanML3D			KIT-ML		
	TWD	TWR	DMSP	R@1↑	R@2↑	R@3↑	R@1↑	R@2↑	R@3↑
1	✗	✗	✗	9.14	13.65	18.72	12.77	21.08	25.30
2	✓	✗	✗	11.91	14.27	20.68	15.78	21.70	29.89
3	✗	✗	✓	11.36	15.18	21.59	15.23	22.53	28.92
4	✓	✓	✓	12.43	15.85	22.17	16.43	22.70	31.93
5	✓	✗	✓	12.24	15.87	22.46	16.99	23.13	31.08
6	✓	✓	✓	14.02	17.58	25.51	18.31	24.82	34.46

Table 3: **Main ablation studies of the proposed key modules on text-to-motion retrieval.** Note that without TWD, TWR is also disabled, since the inverse wavelet transform process is based on the wavelet decomposition process.

5.3.2 FREQUENCY DOMAIN TRANSFORM METHOD

In Table 4, we compare the performance of different frequency domain transform methods, including learnable convolutions, FFT, DCT, DWT, and SWT. Correspondingly, we also adopt parameter-matched convolutions, inverse FFT, DCT, DWT, and SWT for reconstruction in TWR. For fair comparison, we maintain the same number of Transformer layers to encode motions. In the baseline method (Row 1), no transform is used.

Row	Method	HumanML3D			KIT-ML		
		R@1↑	R@2↑	R@3↑	R@1↑	R@2↑	R@3↑
1	N.A.	11.36	15.18	21.59	15.23	22.53	28.92
2	FFT	12.10	15.44	22.07	15.98	23.21	30.62
3	DCT	12.69	16.10	22.84	16.17	23.96	31.15
4	Conv	12.86	15.97	23.40	16.63	24.06	31.08
5	DWT	13.27	16.12	23.91	16.73	23.99	32.59
5	SWT	14.02	17.58	25.51	18.31	24.82	34.46

Table 4: **Ablation study on the frequency domain transform method.** N.A. denotes do not apply transforms.

As shown in Table 4, SWT notably outperforms all other settings across all metrics. Although surpassing the baseline, FFT performs worst due to its global frequency representation, which struggles to capture the essential local temporal dynamics. Convolutions and DCT lead to inferior results compared to DWT and SWT,

![Figure 5: A comparison of human action generation results. The figure shows a sequence of six images for the query 'a person balances on one foot while moving their other, and then switches'. The first image, highlighted with a green border, is the result from WaMo (Ours) and shows a person balancing on one foot. The subsequent images show results from other methods: MoPa (a man is alternating hopping on one foot), TMR (a man leans forward slightly, stands on one foot, and kicks his other foot out behind him quickly), and two additional unlabeled results (a person kicks with their right leg twice, and then once with their left; a person moves their right foot around while balancing on their left foot; and a person is stretching his right leg).](d4e9f8f6bf5d7853ecae9c9633900af1_img.jpg)

Figure 5: A comparison of human action generation results. The figure shows a sequence of six images for the query 'a person balances on one foot while moving their other, and then switches'. The first image, highlighted with a green border, is the result from WaMo (Ours) and shows a person balancing on one foot. The subsequent images show results from other methods: MoPa (a man is alternating hopping on one foot), TMR (a man leans forward slightly, stands on one foot, and kicks his other foot out behind him quickly), and two additional unlabeled results (a person kicks with their right leg twice, and then once with their left; a person moves their right foot around while balancing on their left foot; and a person is stretching his right leg).

Figure 5: Visualization comparisons of fine-grained text-to-motion retrieval between our method, MoPa (Yu et al., 2024), and TMR (Petrovich et al., 2023) on HumanML3D. Top-2 retrieval results are shown from left to right. The ground-truth motion is highlighted by a green box.

primarily due to the inadequate consideration of low-frequency and high-frequency components, respectively. DWT further shows moderate improvements but suffers from shift-variant decomposition, disrupting the original temporal structure. In contrast, SWT can capture multi-resolution motion semantics while preserving the temporal structure, thus leading to the best performance.

5.3.3 DECOMPOSITION LEVEL

In Table 5, we investigate the influence of the decomposition level S on retrieval performance and reconstruction quality. It is found that the performance consistently increases from $S=1$ to $S=3$ across all retrieval metrics. **FID also decreases significantly, indicating less information loss.** It demonstrates that deeper decomposition captures richer temporal hierarchies. However, the penalty to excessive decomposition may introduce precise matching with texts and motion

5.3.4 HYPERPARAMETERS λ_q AND λ_s

In Figure 4, we show the sensitivity of our model to group number λ_g and shuffle ratio λ_s . Optimal results are achieved when $\lambda_g=16$ and $\lambda_s=0.25$ on both datasets. Particularly, the performances are stable and exhibit limited fluctuations across various λ_g and λ_s choices, indicating the robustness and insensitivity of our method to different hyperparameters.

5.3.5 INTRA- AND INTER-FREQUENCY ATTENTION

In Table 6, we ablate the contribution of the Intra- or Inter-Frequency Attention. Note that we maintain the same number of Transformer layers in all experiments to ensure fair comparison. It is found that solely intra-frequency attention outperforms inter-frequency (Row 1 and 2) due to its capture of fine-grained motion semantics across diverse temporal scales. Moreover, their improvements since the cross-frequency fe-

S	HumanML3D				KIT-ML			
	R@1↑	R@2↑	R@3↑	FID↓	R@1↑	R@2↑	R@3↑	FID↓
1	11.93	15.75	22.21	0.049	15.66	22.85	30.26	0.152
2	12.81	16.39	23.16	0.032	16.99	23.73	30.96	0.127
3	14.02	17.58	25.51	0.018	18.31	24.82	34.46	0.085
4	13.84	17.20	24.52	0.026	18.19	24.06	31.33	0.112

Table 5: Ablations on decomposition level S .

![Figure 10: Two line graphs showing the effect of hyperparameters on R@1. The left graph shows R@1 vs Group Number K1 (4 to 224), and the right graph shows R@1 vs Shuffle Ratio K2 (0.15 to 0.75). Both graphs compare KiT-ML (blue line with circles) and HumanML3D (red line with circles).](ca5f2304ffe22946414ffab54cd3fd93_img.jpg)

Figure 10 consists of two line graphs. The left graph plots R@1 (y-axis, 15 to 18) against Group Number K_1 (x-axis, 4 to 224). The right graph plots R@1 (y-axis, 14 to 18) against Shuffle Ratio K_2 (x-axis, 0.15 to 0.75). Both graphs compare KiT-ML (blue line with circles) and HumanML3D (red line with circles). In both cases, KiT-ML maintains a high R@1 (around 17.5-18.0), while HumanML3D's performance is significantly lower and more stable (around 14.0-14.2).

Hyperparameter	Value	KiT-ML R@1	HumanML3D R@1

K_1 (Group Number)	4	~17.5	~14.0
	8	~18.0	~14.1
	16	~18.2	~14.1
	32	~17.5	~14.0
	64	~17.3	~13.9
	224	~17.5	~13.8
K_2 (Shuffle Ratio)	0.15	~18.0	~14.1
	0.25	~18.2	~14.2
	0.5	~17.8	~14.1
	0.75	~17.5	~13.9

Figure 10: Two line graphs showing the effect of hyperparameters on R@1. The left graph shows R@1 vs Group Number K1 (4 to 224), and the right graph shows R@1 vs Shuffle Ratio K2 (0.15 to 0.75). Both graphs compare KiT-ML (blue line with circles) and HumanML3D (red line with circles).

Figure 4: Ablation study on the group number λ_g and shuffle ratio λ_s .

Row	Attention		HumanML3D			KIT-ML		
	<i>intra</i>	<i>inter</i>	R@1%	R@2%	R@3%	R@1%	R@2%	R@3%
1	✓	✗	13.17	16.12	24.02	17.71	22.97	31.08
2	✗	✓	12.82	15.74	23.36	17.11	22.71	30.12
3	✓	✓	14.02	17.58	25.51	18.31	24.82	34.46

Table 6: Ablations on Intra- or Inter-Frequency Attention. joint application (Row 3) yields further performance im- sion integrates complementary information.

5.3.6 TRAJECTORY WAVELET RECONSTRUCTION

In Table 7, we demonstrate the effectiveness of Trajectory Wavelet Reconstruction. Both intra- and inter-frequency reconstruction result in improvements on all metrics (Row 1-3), since these reconstruction constraints enforce the motion encoder to extract more relevant frequency-localized details and holistic motion semantics corresponding to the text. Their combination further leads to the best performance, indicating the complementary nature of both localized and holistic processing. It verifies the reasonability of proposed methods.

Row	Reconstruction		HumanML3D			KIT-ML		
	<i>intra</i>	<i>inter</i>	R@1↑	R@2↑	R@3↑	R@1↑	R@2↑	R@3↑
1	✗	✗	12.43	15.85	22.17	16.43	22.70	31.93
2	✓	✗	13.27	16.42	23.94	17.47	23.46	33.18
3	✗	✓	13.11	16.21	23.89	17.22	23.14	32.97
4	✓	✓	14.02	17.58	25.51	18.31	24.82	34.46

Table 7: Ablations on Trajectory Wavelet Reconstruction. Their combination further leads to the best performance, indicating the complementary nature of both localized and holistic processing. It verifies the reasonability of proposed methods.

5.4 DISORDERED MOTION SEQUENCE PREDICTION

To quantitatively verify the effectiveness of the Disordered Motion Sequence Prediction (DMSP) module, we compare our method with two variants: one baseline without sequence shuffling, and the other one replacing DMSP with the 2D video shuffling technique (Lee et al., 2017). As shown in Table 8, the video shuffling method significantly underperforms our DMSP and even falls behind the baseline. It indicates that preserving local continuity is critical for learning motion representations.

Setting	HumanML3D			KIT-ML		
	R@1↑	R@2↑	R@3↑	R@1↑	R@2↑	R@3↑
WaMo w/o shuffling	12.24	15.87	22.46	16.99	23.13	31.08
WaMo w/ video shuffling	11.56	15.42	21.55	15.03	22.59	28.89
WaMo w/ DMSP (Ours)	14.02	17.58	25.51	18.31	24.82	34.46

Table 8: Ablation study on the Disordered Motion Sequence Prediction module.

5.5 QUALITATIVE RESULTS

We present a comparative visualization of fine-grained text-to-motion retrieval among our method, MoPa, and TMR in Figure 5. While baseline methods retrieve basic actions like “move the foot”, they fail to capture specific semantic details such as “switch the foot”. In contrast, our method precisely retrieves motions that fully align with the complex textual description depicting fine-grained actions. It demonstrates our method’s effectiveness in aligning intricate motions with texts.

6 CONCLUSION

We introduce WaMo, a novel wavelet-based framework that significantly enhances Text-Motion Retrieval (TMR) through comprehensive multi-frequency trajectory analysis. WaMo addresses key challenges in TMR through three key modules. Trajectory Wavelet Decomposition fully captures multi-scale frequency information. Trajectory Wavelet Reconstruction ensures the preservation of spatial-temporal information. Disordered Motion Sequence Prediction improves the learning of temporal structures. Experimental results on HumanML3D and KIT-ML datasets indicate that WaMo outperforms existing methods by substantial margins.

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Rest of paper (reference and Appendix) is removed.